

Problem 15.18

Nancy has \$10,000 in a bank account earning a nominal annual rate of 6% compounded monthly. Calculate the amount that she can withdraw at the end of each month from the account if she wants to have zero in the account after 12 months.

Solution

The nominal annual interest rate compounded monthly is 6%, so the monthly interest rate is

$$j = \frac{0.06}{12} = 0.005.$$

Let X be the monthly withdrawal. Since the account is depleted after 12 months, the initial balance is equal to the present value of the 12 monthly withdrawals:

$$10000 = Xa_{\overline{12}|j}.$$

Using

$$a_{\overline{n}|j} = \frac{1 - (1 + j)^{-n}}{j},$$

we get

$$10000 = X \left(\frac{1 - (1.005)^{-12}}{0.005} \right).$$

Solving for X ,

$$\begin{aligned} X &= \frac{10000}{\frac{1 - (1.005)^{-12}}{0.005}} \\ &\approx 860.6643. \end{aligned}$$

Hence, Nancy can withdraw

$$\boxed{\$860.66}$$

at the end of each month.